**Generative AI paper**

# Effects of Block-Based Programming Instruction on Computational Thinking in School Students: A Meta-Analysis

**Society for Educational Data Analysis (SEDA)**

*(The Japanese version of this paper, with identical content, is available at https://raw.githubusercontent.com/k518-2026/MetaAnalysisToWP/main/reports/2026-10-11-info-block-programming-ct/paper-ja.pdf. Published October 11, 2026.)*

***Abstract*** Fostering computational thinking skills in school education is increasingly vital for preparing young learners to solve complex problems in digital environments. Using an automated workflow in which every extracted number was verified against the full text, 9 open-access studies ($N$ = 806) were pooled with a random-effects model. The pooled effect was $g$ = 1.47, 95% CI [0.87, 2.06], with $I^2$ = 87.4%. Block-based programming instruction can effectively foster computational thinking across primary and lower secondary classrooms compared to both passive and active alternatives.

***Keywords*** meta-analysis, block-based programming, visual programming, computational thinking, elementary education, middle school

## 1. Introduction

Computational thinking has emerged as an essential skill within primary and lower secondary school curricula. As schools aim to prepare young learners for problem-solving in modern technology-rich societies, introducing algorithmic thinking and programming principles during elementary and middle school grades has become a major educational priority. However, implementing programming instruction in early grades requires pedagogical approaches that are developmentally appropriate for novice students.

Visual and block-based programming environments, such as Scratch, allow students to assemble programs using graphical blocks rather than typing text-based code. The included primary studies evaluated various block-based implementations across elementary and middle school grades, ranging from standalone visual programming activities [1], [2] to visual coding integrated into science curricula [3], [4]. Other studies investigated block-based curricula that transition into text-based languages [5] or focused on structured visual frameworks such as Sense-Reason-Act programming environments [6], [7].

Although numerous empirical evaluations have examined visual programming, their findings are distributed across distinct instructional settings, comparison conditions, and school levels. Prior studies have compared block-based instruction to passive baselines [1], [2], [3], [7] as well as active instructional alternatives [4], [6], [8], [9]. A quantitative synthesis is therefore useful to combine these diverse findings into a single pooled estimate. The purpose of this meta-analysis is to evaluate the effects of block-based programming instruction on computational thinking in school students, examining overall efficacy as well as potential variations across school levels and comparison types.

## 2. Method

### 2.1 Literature Search

Searches were run on October 11, 2026 in two open bibliographic sources. OpenAlex was searched in the title and abstract fields with the query “(Scratch OR "block-based programming" OR "visual programming" OR "block-based coding") AND ("computational thinking" OR "problem solving" OR programming) AND (elementary OR primary OR "middle school" OR "junior high" OR secondary OR "K-12" OR grade OR children OR pupils)”, restricted to open-access journal articles published in 2012 or later and classified in the Education or Developmental and Educational Psychology subfields. J-STAGE, the Japanese journal platform, was searched in the abstract field with the keywords “Scratch 小学校”, “ビジュアルプログラミング 授業” (2012 or later) to include studies published in Japan.

### 2.2 Eligibility Criteria

Studies were eligible if they (a) sampled students in primary or secondary school (K-12); Primary and lower secondary samples are preferred; upper secondary samples are acceptable; Exclude university, adult, and teacher/pre-service teacher samples; (b) examined block-based or visual programming instruction (e.g., Scratch, ScratchJr, Code.org, Blockly); (c) compared it with regular lessons, no programming instruction, or text-based/alternative instruction; (d) measured a computational thinking, programming, or problem-solving test; and (e) reported statistics sufficient to compute a standardized mean difference. Reviews, qualitative studies, single-case designs, and single-group pre-post designs were excluded.

### 2.3 Screening and Data Extraction

Screening and data extraction were automated. A large language model (gemini-3.8-flash, gemini-3.6-flash,

gemini-3.7-flash) rated each title and abstract against the eligibility criteria. Full texts (PDF) of the highest-rated records were retrieved and converted to plain text, and the model extracted study characteristics and summary statistics from them. Every extracted number was then checked by software against the full text, and a study was excluded from the synthesis if any extracted number could not be found verbatim. The model also classified the comparison condition of each study as passive (no intervention, a waitlist, or regular lessons) or active (an alternative intervention aimed at the same outcome) and quoted the passage supporting the classification; a classification whose quotation could not be found in the full text was recorded as unclear. No human reviewer verified the screening or the extraction. When more than 10 studies qualified, the 10 with the highest screening ratings were retained.

### 2.4 Effect Sizes

Standardized mean differences were computed as Hedges' g [10] from post-test means, standard deviations, and group sizes; when these were unavailable, from independent-samples t statistics, from F statistics with one numerator degree of freedom, or from reported d or g values. Positive values indicate higher scores in the intervention group. One effect size per study was used.

### 2.5 Statistical Analysis

Effect sizes were pooled with a random-effects model in which the between-study variance ($\tau^2$) was estimated with the DerSimonian–Laird method [11]. Confidence intervals and tests of the pooled effect used the Hartung–Knapp–Sidik–Jonkman (HKSJ) adjustment, which is recommended when the number of studies is small [12], [13]. Heterogeneity was assessed with Q, $I^2$ [14], $\tau^2$, and a 95% prediction interval. Subgroup analyses by school level (elementary: grades 1–6; lower secondary: grades 7–9) were conducted when each level included at least two studies. A further subgroup analysis compared studies with a passive comparison condition with those with an active comparison condition, again requiring at least two studies per type; studies whose comparison type was unclear were left out of this comparison only. Robustness was examined with leave-one-out analysis, and Egger's regression test [15] was planned only when at least 10 studies were available. Effect magnitudes were interpreted with conventional benchmarks [16]. Analyses followed standard formulas [17] implemented in open-source JavaScript code (https://github.com/k518-2026/MetaAnalysisToWP).

## 3. Results

### 3.1 Study Selection

The searches returned 200 records from OpenAlex (of 612 matching the query) and 37 from J-STAGE, for 233 unique records. Of the 200 records with abstracts that were screened, 31 were rated as potentially eligible. Full texts of 30 records were examined: 8 could not be retrieved or read, 6 did not meet the eligibility criteria, 5 did not report sufficient statistics, 2 contained extracted numbers that could not be verified in the text, and 0 yielded implausible values. In total, 9 studies were included in the meta-analysis.

### 3.2 Study Characteristics

Table 1 summarizes the country, grade, design, sample size, and effect size of each included study [1], [2], [3], [4], [5], [6], [7], [8], [9]. The studies comprised 806 participants in total and were conducted in Turkey, Portugal, China, Netherlands. By school level, 6 studies sampled elementary (grades 1–6) students, 3 studies sampled lower secondary (grades 7–9) students. Sample sizes ranged from 30 to 200, and individual effect sizes ranged from $g = 0.41$ to $g = 2.67$.

**Table 1.** Overview of the Included Studies

| Study | Country | Grade | Design | *N* | *g* [95% CI] |
|---|---|---|---|---|---|
| Namlı and Aybek [1] | Turkey | Grade 5 | randomized controlled trial | 58 | 0.92 [0.39, 1.46] |
| Oluk et al. [8] | Turkey | Grade 5 | quasi-experimental | 62 | 2.67 [1.99, 3.35] |
| Piedade and Dorotea [2] | Portugal | Grade 4 | quasi-experimental | 189 | 1.16 [0.85, 1.46] |
| Sungur and Özkan [3] | Turkey | Grade 6 | quasi-experimental | 51 | 1.67 [1.04, 2.31] |
| Wang et al. [5] | China | Grades 3-6 | randomized controlled trial | 200 | 1.60 [1.28, 1.92] |
| Çetin and OTU [9] | Turkey | Grade 6 | quasi-experimental | 66 | 2.42 [1.79, 3.05] |
| ÖZTÜRK and Arabacıoğlu [4] | Turkey | Grade 6 | quasi-experimental | 41 | 0.65 [0.03, 1.27] |
| Fanchamps et al. [6] | Netherlands | Grade 5 | randomized controlled trial | 30 | 2.05 [1.18, 2.92] |
| Fanchamps et al. [7] | Netherlands | Grades 5-6 | randomized controlled trial | 109 | 0.41 [0.03, 0.79] |

*Note.* CI = confidence interval. *g* = Hedges' g (positive values favor the intervention). Grades and designs were extracted automatically from the full texts.

Table 2 lists the intervention, comparison condition, and outcome measure of each study. All 9 studies compared an intervention group with a comparison group on a post-test.

**Table 2.** Interventions, Comparison Conditions, and Outcome Measures

| Study | Intervention | Comparison | Outcome |
|---|---|---|---|
| Namlı and Aybek [1] | block-based programming activities | regular curriculum lessons with no interventions [passive] | International Informatics and Computational Thinking Activity Task Test |
| Oluk et al. [8] | Scratch visual programming instruction | regular curriculum-based algorithm instruction [active] | Algorithm Development Achievement Test |
| Piedade and Dorotea [2] | Scratch-based programming activities | no programming instruction [passive] | Beginners Computational Thinking Test score |
| Sungur and Özkan [3] | Scratch-integrated science instruction | standard science curriculum [passive] | Computational Thinking Test (CTt) |
| Wang et al. [5] | 24-week coding curriculum using Scratch and Python | regular school curriculum with basic ICT literacy [passive] | computational thinking assessment |
| Çetin and OTU [9] | Scratch block-based programming instruction | Python text-based programming instruction [active] | Programming Achievement Test (PAT) |
| ÖZTÜRK and Arabacıoğlu [4] | Scratch block-based coding activities integrated into science lessons | unplugged algorithmic explanation activities integrated into science lessons [active] | Computational Thinking Scale |
| Fanchamps et al. [6] | Visual programming with Sense-Reason-Act (SRA) concepts | Visual programming using linear, sequential programming without SRA concepts [active] | Computational Thinking Test (CTt) |
| Fanchamps et al. [7] | visual SRA-programming instruction using Bomberbot | regular curriculum without programming instruction [passive] | Computational Thinking test (CTt) |

*Note.* Extracted and summarized automatically from the full texts. [passive] = no intervention or regular lessons; [active] = an alternative intervention aimed at the same outcome. Unclear cases carry no label.

### 3.3 Overall Effect

Fig. 1 shows the forest plot. Each square marks the effect size of one study, with its size proportional to the study's weight, and each horizontal line marks the 95% confidence interval; the diamond at the bottom is the pooled estimate. 9 of the 9 studies had positive effect sizes, and the confidence intervals of 9 studies did not include zero. The study with the largest weight was Piedade and Dorotea [2] (12.6%).

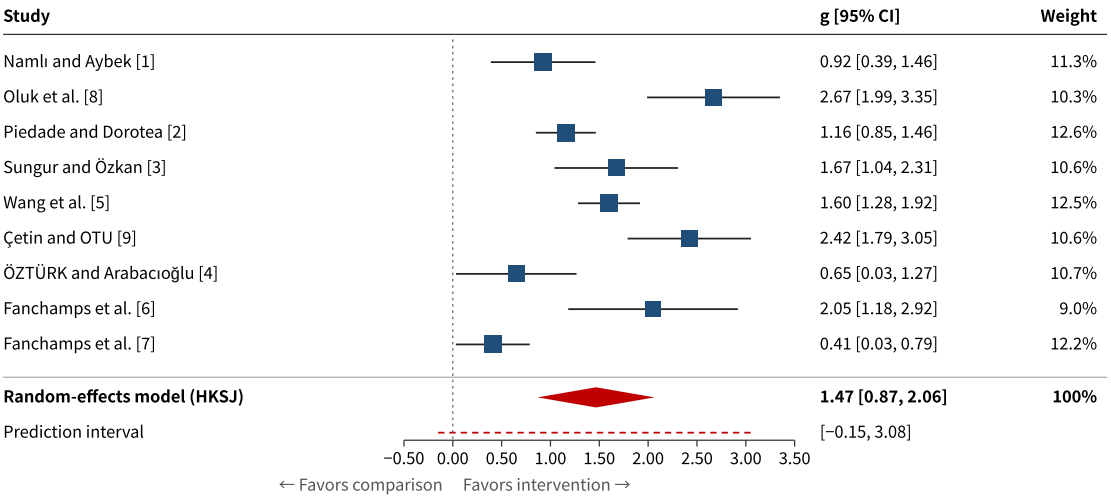



**Figure 1.** Forest plot of individual and pooled effect sizes

Squares show study effect sizes (size proportional to weight), lines show 95% confidence intervals, the diamond shows the pooled estimate, and the dashed line shows the prediction interval.

The random-effects model yielded a pooled effect of $g$ = 1.47, 95% CI [0.87, 2.06], $t(8)$ = 5.66, p < .001, which corresponds to a large effect by conventional benchmarks. The fixed-effect estimate was $g$ = 1.30. Table 3 summarizes these estimates.

### 3.4 Heterogeneity

Heterogeneity was considerable, $Q(8)$ = 63.62, p < .001, $I^2$ = 87.4%, $\tau^2$ = 0.413. The 95% prediction interval ranged from −0.15 to 3.08.

### 3.5 Subgroup and Sensitivity Analyses

For elementary (grades 1–6) samples, $g$ = 1.41, 95% CI [0.57, 2.25] ($k$ = 6); for lower secondary (grades 7–9) samples, $g$ = 1.58, 95% CI [−0.63, 3.79] ($k$ = 3). The difference between school levels was not statistically significant, $Q(1)$ = 0.08, $p$ = .783. By type of comparison condition, passive comparison (no intervention or regular lessons): $g$ = 1.14, 95% CI [0.50, 1.78] ($k$ = 5); active comparison (an alternative intervention): $g$ = 1.94, 95% CI [0.49, 3.39] ($k$ = 4). The difference between comparison types was not statistically significant, $Q(1)$ = 2.44, $p$ = .118; this comparison is exploratory and rests on few studies. Leave-one-out estimates ranged from 1.32 to 1.60. Publication bias was not tested because fewer than 10 studies were included. The subgroup estimates are also shown in Table 3.

**Table 3.** Results of the Meta-Analysis

| Model | *k* | *N* | *g* | 95% CI | *p* | $I^2$ (%) | $\tau^2$ |
|---|---|---|---|---|---|---|---|
| Random effects (HKSJ) | 9 | 806 | 1.47 | [0.87, 2.06] | < .001 | 87.4 | 0.413 |
| Fixed effect | 9 | 806 | 1.30 | [1.14, 1.45] | < .001 | — | — |
| elementary (grades 1–6) | 6 | 648 | 1.41 | [0.57, 2.25] | .008 | 89.0 | 0.398 |
| lower secondary (grades 7–9) | 3 | 158 | 1.58 | [−0.63, 3.79] | .091 | 87.2 | 0.697 |
| passive comparison (no intervention or regular lessons) | 5 | 607 | 1.14 | [0.50, 1.78] | .008 | 84.5 | 0.225 |
| active comparison (an alternative intervention) | 4 | 199 | 1.94 | [0.49, 3.39] | .024 | 87.2 | 0.837 |

*Note.* $k$ = number of studies. HKSJ = Hartung–Knapp–Sidik–Jonkman confidence interval. 95% prediction interval [−0.15, 3.08].

## 4. Discussion

The synthesis revealed a statistically significant, large overall effect of block-based programming instruction on computational thinking outcomes ($g$ = 1.47, 95% CI [0.87, 2.06], p < .001; Table 3). Across the included studies, effect sizes consistently favored the intervention over comparison groups (Fig. 1). Substantial heterogeneity was present across the sample of studies ($Q(8)$ = 63.62, p < .001, I2 = 87.4%, tau2 = 0.413; Table 3), accompanied by a wide prediction interval of [−0.15, 3.08]. A sensitivity analysis examining leave-one-out estimates demonstrated that the pooled effect remained relatively stable, ranging from 1.32 to 1.60.

Subgroup analyses were conducted to explore school level and comparison type, although these comparisons are exploratory and based on few studies. In the analysis by school level (Table 3), elementary samples showed a large effect ($k$ = 6, $g$ = 1.41, 95% CI [0.57, 2.25]), which did not differ significantly from the middle school samples ($k$ = 3, $g$ = 1.58, 95% CI [−0.63, 3.79]; $Q(1)$ = 0.08, $p$ = .783). Similarly, when examining comparison types, active comparisons yielded an estimate of $g$ = 1.94 (95% CI [0.49, 3.39], $k$ = 4), whereas passive comparisons yielded $g$ = 1.14 (95% CI [0.50, 1.78], $k$ = 5), with no statistically significant subgroup difference ($Q(1)$ = 2.44, $p$ = .118). Because of the small number of studies in these subsets, these subgroup differences remain exploratory and cannot establish definitive causes.

An inspection of the individual studies summarized in Table 1 and Table 2 reveals variations across instructional designs and control conditions. Large effects were reported in elementary cohorts when Scratch was contrasted with regular algorithm instruction ($g$ = 2.67 [8]) and when visual Sense-Reason-Act concepts were contrasted with linear visual programming ($g$ = 2.05 [6]). Moderate to large effects occurred in elementary evaluations comparing block-based instruction with regular curriculum controls ($g$ = 0.92 [1]; $g$ = 1.16 [2]; $g$ = 1.60 [5]; $g$ = 0.41 [7]). In middle school cohorts, block-based learning produced substantial gains over text-based Python instruction ($g$ = 2.42 [9]) and regular science lessons ($g$ = 1.67 [3]), while comparison against unplugged algorithmic instruction in science yielded a smaller effect ($g$ = 0.65 [4]).

For teachers of grades 1-9, these findings indicate that block-based programming environments can serve as an effective means to develop student computational thinking. Visual environments may be implemented as dedicated computing units or integrated within subject areas such as science. Furthermore, teachers working with novice learners may find visual block platforms beneficial when introducing algorithmic principles before or alongside text-based or unplugged instructional alternatives.

## 5. Conclusion

In conclusion, this meta-analysis of 9 studies indicates that block-based programming instruction has a large positive effect on computational thinking in school students compared to both passive and active control conditions. Although the small number of studies and substantial heterogeneity require cautious

interpretation, visual block-based tools provide a promising approach for cultivating algorithmic and computational skills across grades 1 through 9.

This meta-analysis has several limitations. The synthesis is based on a small total evidence base of 9 studies encompassing 806 participants, which restricts statistical power, prevented the assessment of publication bias, and means subgroup findings must be regarded as exploratory. Furthermore, the underlying data were extracted automatically, which may limit the precision of nuanced study characteristics.

In addition, the included studies displayed variability in study designs and assessment instruments. The evidence base comprises both randomized controlled trials [1], [5], [6], [7] and quasi-experimental designs [2], [3], [4], [8], [9]. Outcome measures also varied from standardized computational thinking tests to specific programming achievement tests and self-report scales, contributing to the substantial heterogeneity observed across findings.

In addition, screening and data extraction were automated with a language model, and extracted numbers were verified only by matching them against the full text. A number can be present in the text yet belong to a different group or measure. Only open-access studies whose full texts could be retrieved were included; subscription-only and unpublished studies were not.

## Author notes

This is a “Generative AI paper” produced automatically by the Society for Educational Data Analysis (SEDA) with generative AI. Large language models (gemini-3.8-flash, gemini-3.6-flash, gemini-3.7-flash) were used to screen records, extract data, and draft the Introduction, Discussion, and Conclusion; all statistics were computed by software. The paper has not been peer reviewed.

Data and code are available at https://github.com/k518-2026/MetaAnalysisToWP. © 2026 Society for Educational Data Analysis (SEDA).

## References

References marked with an asterisk indicate studies included in the meta-analysis.

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